SECURITY DOOR LOCK SYSTEM

Biometric access + CCTV + web-based monitoring | Assumption University

A smarter way to monitor who enters, when, and what happens

Senior Project 1: prototype

Senior Project 2: integration

The problem we are solving

Access control and visual monitoring are often managed separately.

WHO entered?

A door lock may grant access without providing a convenient central record.

WHEN did it happen?

Separate logs make it harder to review entry and exit events.

WHAT happened?

CCTV footage is useful but disconnected from identity records.

Our proposed solution

One locally hosted dashboard combining biometric records and camera monitoring.

IDENTIFY

Fingerprint / supported biometrics

RECORD

User + time + access outcome

MONITOR

CCTV live view / footage

REVIEW

Unified web dashboard

Goal: a practical system for a controlled entrance in an AU building.

Project objectives

From isolated devices to a usable campus security interface.

- Connect the Comet FZ05 biometric device to the local computer/server.
- Connect a CCTV camera to the network for website monitoring.
- Use a repurposed Dell OptiPlex as the local Linux server.
- Display user records, access events and CCTV in one custom website.

System components

Hardware and software identified in the project report.

Comet FZ05

Biometric access / attendance device

8MP 4K CCTV

Video monitoring camera

Dell OptiPlex

Reused local server computer

Ethernet + Wi-Fi

Local network connectivity

Ubuntu Linux

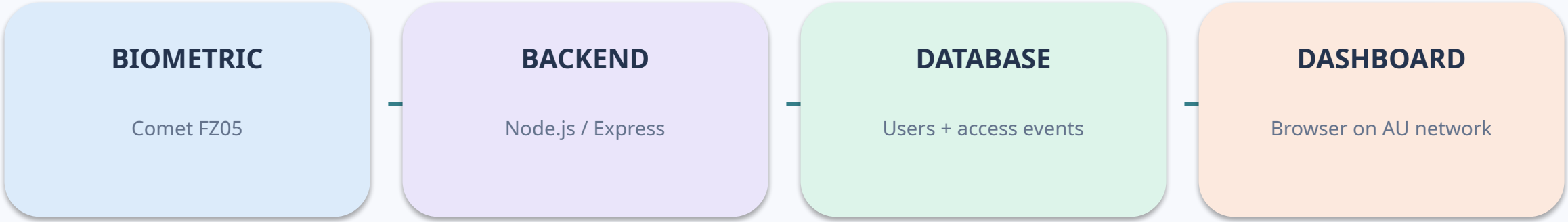
Server operating environment

Custom website

HTML, CSS and JavaScript interface

Architecture: target end-to-end system

Planned final integration; arrows represent intended data flow.



CCTV camera → video service / stream → CCTV page in the same dashboard

Senior Project 1: work completed

Built the website prototype and prepared the Linux hosting environment.

WEB INTERFACE

Five website pages designed and tested with sample data.

Dashboard • Users • Access Logs • CCTV • Settings

LOCAL SERVER

Dell OptiPlex running Ubuntu Linux.

Website accessible through a browser on the local AU network (as described in the report).

Website structure

Five views support day-to-day administration.

Dashboard

Overview / device status

Users

Registered identities

Access Logs

Event history / CSV

CCTV

Video + snapshot

Settings

Network + devices

Dashboard and Users prototype

Screenshots included in the current report.

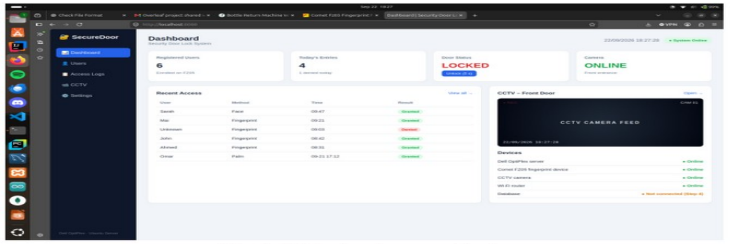


Fig. 8: Website interface screenshot 1.

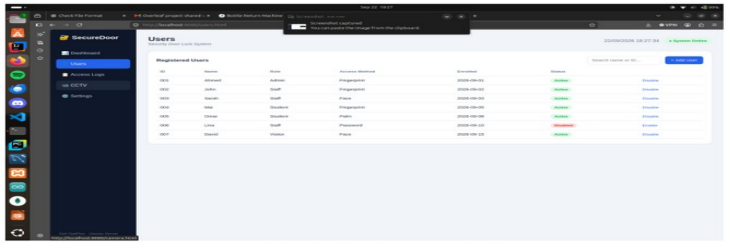


Fig. 9: Website interface screenshot 2.

WHAT IT SHOWS

Dashboard: registered users, daily access activity, door state, CCTV preview and connectivity.

Users: search, add and manage displayed user records.

Currently tested with sample data.

Access Logs and Settings prototype

Screenshots included in the current report.

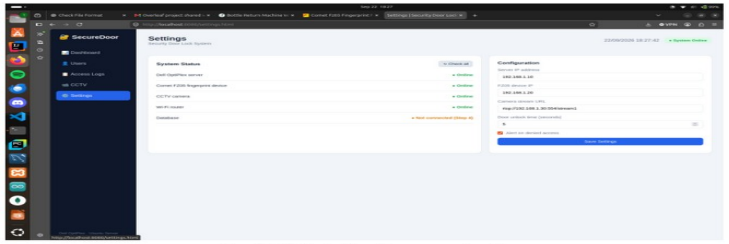


Fig. 10: Website interface screenshot 3.

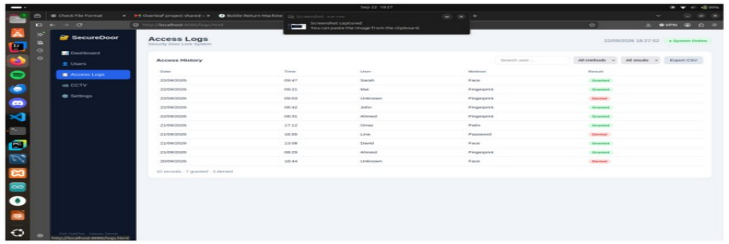


Fig. 11: Website interface screenshot 4.

WHAT IT SHOWS

Access Logs: time, user, method, outcome, filters and CSV export.

Settings: device status and configuration fields.

Hardware-backed functionality remains to be integrated.

CCTV streaming — implementation & live demo

Jibrael • Ubuntu Linux • FFmpeg • RTSP

01 VIDEO SOURCE
Camera / test feed

→

02 RTSP STREAM
Local RTSP service

→

03 PROCESSING
FFmpeg pipeline

→

04 WEBSITE
SecureDoor CCTV page

▶️ LIVE CCTV DEMONSTRATION

Play your recorded CCTV demo video here. Show the Ubuntu streaming setup and the camera view in the SecureDoor website.

Current status: streaming pipeline tested; confirm stable browser playback before final deployment.

Why Ubuntu and the Dell OptiPlex?

A practical local-server choice for a student prototype.

REUSE

Existing desktop hardware reduces new equipment needs.

CONTROL

Linux supports permissions, firewall tools and remote administration.

HOSTING

One local machine can host the website and supporting services.

Current status at a glance

A transparent view of what is implemented versus what is still planned.

Website frontend	BUILT / SAMPLE DATA
Ubuntu local hosting	SET UP / REPORTED
CCTV streaming	IN TESTING
Database + API	TO BUILD
Comet FZ05 integration	TO CONNECT
End-to-end access tests	TO VALIDATE

Senior Project 2: implementation plan

Move from a frontend demonstration to an integrated working prototype.

1. DATA

Database schema + storage

2. API

Node.js / Express services

3. DEVICE

Read FZ05 user / access records

4. VIDEO

Stabilize browser CCTV view

5. TEST

Validate real scenarios

Database design

Proposed logical entities; final fields depend on the device interface.

USERS

user_id • name • role

DEVICES

device_id • type • status

ACCESS EVENTS

event_id • user_id • time • result

CAMERA EVENTS

camera_id • time • reference

Design principles: unique identifiers, event timestamps, controlled access and retention rules.

Fingerprint / biometric device integration

Comet FZ05 must provide usable records to the backend.

ENROLL

Register approved user

SCAN

Device identifies user

SYNC

Import / receive device record

STORE

Write event to database

DISPLAY

Update access logs

Check the FZ05 export / USB / vendor interface first. Do not assume a live API exists.

End-to-end operation at the classroom entrance

Target workflow for a real-world pilot installation.

ARRIVE

Authorized person approaches

AUTHENTICATE

Fingerprint or supported method

DECIDE

Grant / deny per device logic

LOG

Save user + time + result

REVIEW

Admin checks log + CCTV

Camera records the monitored entrance; access logs and footage can be reviewed from one interface.

Testing and acceptance criteria

A successful demonstration needs more than a working page.

- Registered user: correct identity, timestamp and access result recorded.
- Unknown or unauthorized attempt: correct denial event displayed.
- Camera: stable viewing from an authorized browser on the local network.
- Database: records persist after restart; filters and exports work.
- Security: administrator authentication, access controls and data protection.

Deployment in one AU classroom

The confirmed scope is one AU classroom; installation requires university approval and safeguards.

PILOT ENVIRONMENT

One designated AU classroom entrance.

Dedicated device/network placement.

Named university operator and maintenance contact.

BEFORE REAL USE

Confirm physical door-control compatibility, consent/privacy rules, CCTV retention, network security, reliability and emergency egress.

Get AU facilities / IT authorization.

Roadmap to the finished system

Senior Project 1 → Senior Project 2 → one-classroom AU deployment.

NOW

Website + Linux server + CCTV testing

NEXT

Database + backend + FZ05

THEN

Integration + security testing

GOAL

Working system in one AU classroom

A unified, locally managed access-monitoring system for Assumption University.